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Module- 10\11

Student Performance Prediction System using Machine Learning and GitHub

2. Problem Statement

Educational institutions can use predictive analytics to identify students who may require additional support. This project uses machine learning techniques to analyze student data and predict academic outcomes.

4. Tools and Technologies Used

Python

Pandas

NumPy

Matplotlib

Seaborn

Scikit-learn

Git

GitHub

Jupyter Notebook

5. Task Implementation

Task 1: Create GitHub Repository

Created a new GitHub repository.

Initialized the repository with README.md.

Cloned the repository to the local system.

Maintained project files using Git version control.

Git Commands Used:

```
git init
git add .
git commit -m "Initial Commit"
git branch -M main
git remote add origin <repository_link>
git push -u origin main
```

Task 2: Load and Explore Dataset

Loaded the student dataset using Pandas.

Displayed dataset information.

Checked data types, missing values, and summary statistics.

```
df = pd.read_csv("student_performance.csv")
df.head()
df.info()
df.describe()
```

Task 3: Preprocess Data

Removed unnecessary columns.

Handled missing values.

Checked duplicate records.

Selected relevant features.

Features Used:

Study Hours per Week

Attendance Rate

Previous Score

Target Variable:

Passed (Yes/No)

Task 4: Train Machine Learning Model

Split the dataset into training and testing data.

Trained classification models:

Decision Tree Classifier

Logistic Regression

Evaluated model performance using:

Accuracy Score

Confusion Matrix

Classification Report

```
model.fit(X_train, y_train)
predictions = model.predict(X_test)
```

Task 5: Push Code & Collaborate

Uploaded project files to GitHub.

Used commits to track changes.

Collaborated with team members using branches and pull requests.

Git Commands:

```
git add .
git commit -m "Updated ML Model"
git push origin main
```

6. Results

Successfully predicted student pass/fail status.

Study hours, attendance, and previous scores were important factors.

Model achieved good prediction accuracy.

7. Conclusion

This project demonstrates how Machine Learning and GitHub can be combined to build, manage, and collaborate on a student performance prediction system. The model helps identify students who may need academic support and enables efficient project version control through GitHub.

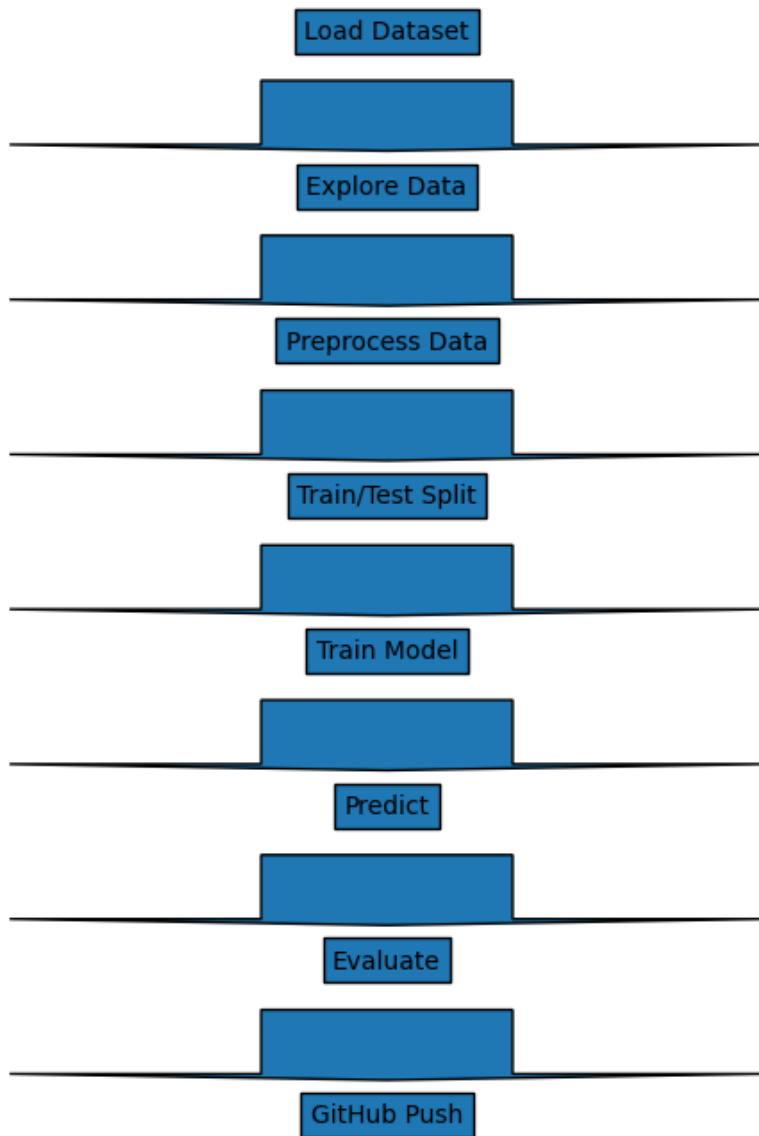
8. Future Scope

Improve accuracy using Random Forest and XGBoost.

Deploy the model using Streamlit.

Add more student-related features.

Enable real-time prediction through a web application.



GitHub Version Control Workflow

